

ELI5 How does a data center make electricity bills go up

Posted by u/gaeyboi69 • 0 points • 88 comments

Why does an ai data center have an effect on the rest of the community's power bills. I understand they take a lot of energy but how does that translate to charge everyone more instead of just charging the data center itself?

Comments

u/TrainDonutBBQ • 1 points • 2025-12-02 21:06 UTC

Wait until you see what EVs do. (and I say this as someone who is driving one).

u/[deleted] • 1 points • 2025-09-29 07:50 UTC

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u/ChuqTas • 1 points • 2025-09-28 13:46 UTC

Every jurisdiction is different, but generally a large customer that requires a large grid connection will pay for the augmentation of the power grid required to support them. It doesn't make everyone else's bills go up. Some people just hate anything AI related and will make things up to encourage others to do the same.

u/LoganDark • 0 points • 2026-03-14 20:30 UTC

Just because you don't pay for the electrical grid itself doesn't mean you don't pay for the electricity. Datacenters reduce electrical supply which drives up prices, and has nothing to do with the grid's capacity to carry current.

u/ChuqTas • 1 points • 2026-03-15 03:37 UTC

That is absolutely not how it works here in Australia, large consumers have to negotiate a PPA with generators. I would think this system is not unique to us but it's possible you're from another country with a different system.

u/Heavy_Direction1547 • 1 points • 2025-09-27 13:18 UTC

In many cases they are not just taking a big share of capacity but require an expensive increase in overall generating capacity, the cost of which is then borne by all users.

u/vercingetafix • 1 points • 2025-09-27 09:45 UTC

This is only true *IF* the electricity utility is not allowed build any new generation.

In practice, of course it can. And in fact it means that prices fall, as the money paid by the data centre

can help cover the fixed costs of the system like wires and transformers.

Recently in Mississippi the utility has been able to invest hundreds of millions into strengthening the grid without having to increase prices - because they got extra revenue from selling power to data centres

u/[deleted] • 1 points • 2025-09-27 06:35 UTC

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u/Can_O_Murica • 1 points • 2025-09-27 05:40 UTC

Hi I work on grid power systems.

Power demand comes and goes. It's expensive and dangerous to run all the generators all the time, so we always run just enough generators to meet demand.

However, the people whose generators don't get used very often still need to make ends meet, so when they *do* get the call, they get paid more to offset the times when they aren't getting paid.

Some of these data centers have the equivalent power load of like 10,000 houses. If you build one and switch it on, suddenly it's like you have a whole new city in a location that really doesn't have power, so we have to run the backup power all the time.

The kicker is that we all share in the cost, because there's no way to track where individual electrons go. They get emptied into the grid, and then they trickle out of the grid into your home. Did your electrons come from the cheap generator, or the expensive one? No one knows, so we all pay the markup.

u/libra00 • 1 points • 2025-09-27 03:05 UTC

If you have 100 units of electricity you can sell and you've been pretty reliably selling 85 of it to various homes and businesses and then someone comes along and wants to buy 50 units, where do you get the extra? You raise the price until everyone else cuts back their usage enough that you can sell that 50. This is supply and demand; if demand increases relative to supply, the price will go up.

u/TheLuo • 1 points • 2025-09-27 00:34 UTC

The energy has to come from somewhere. Those somewhere have limitations on how much they can generate. If the generation limits are reached the energy has to come from somewhere else.

Transmission of electricity from further away costs more. So to fulfill all the demands for energy in this area now costs more. For everyone.